

PROJECT REPORT

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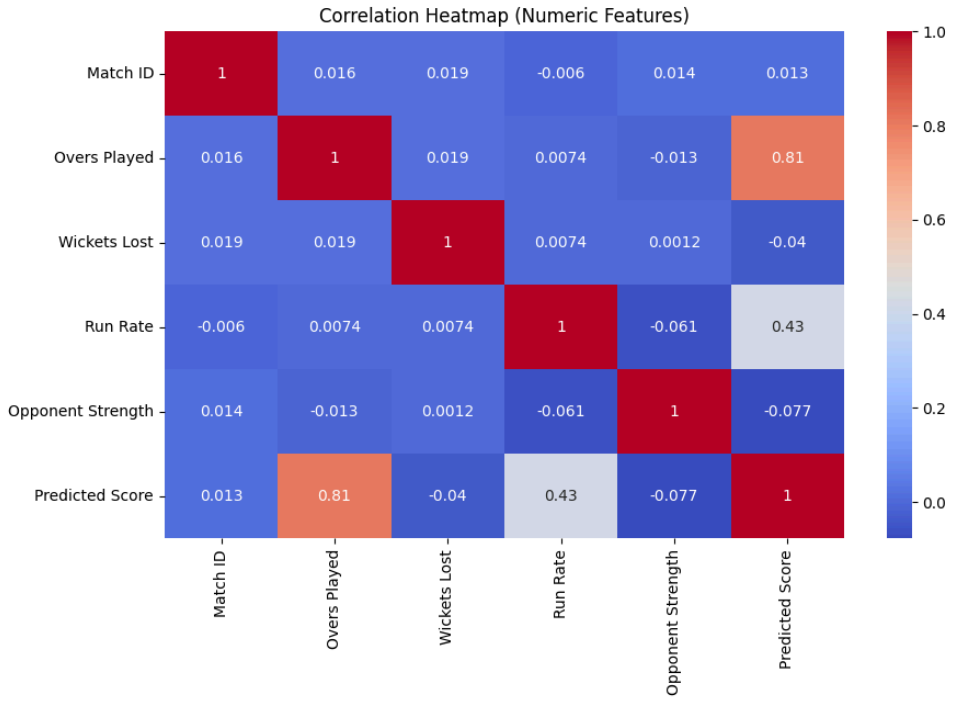
Introduction

The aim of this project is to apply Machine Learning techniques to predict and analyze cricket match scores based on various match related features.By training and testing different regression models this project attempts to identify the most suitable and accurate model for predicting team scores. The motivation behind this work is to develop a data driven system that can assist analysts, coaches and fans in understanding performance patterns and estimating match outcomes more effectively. With the dynamic nature of modern cricket and the increasing availability of match data such predictive models can provide valuable insights and enhance decision making in the field of sports analytics.

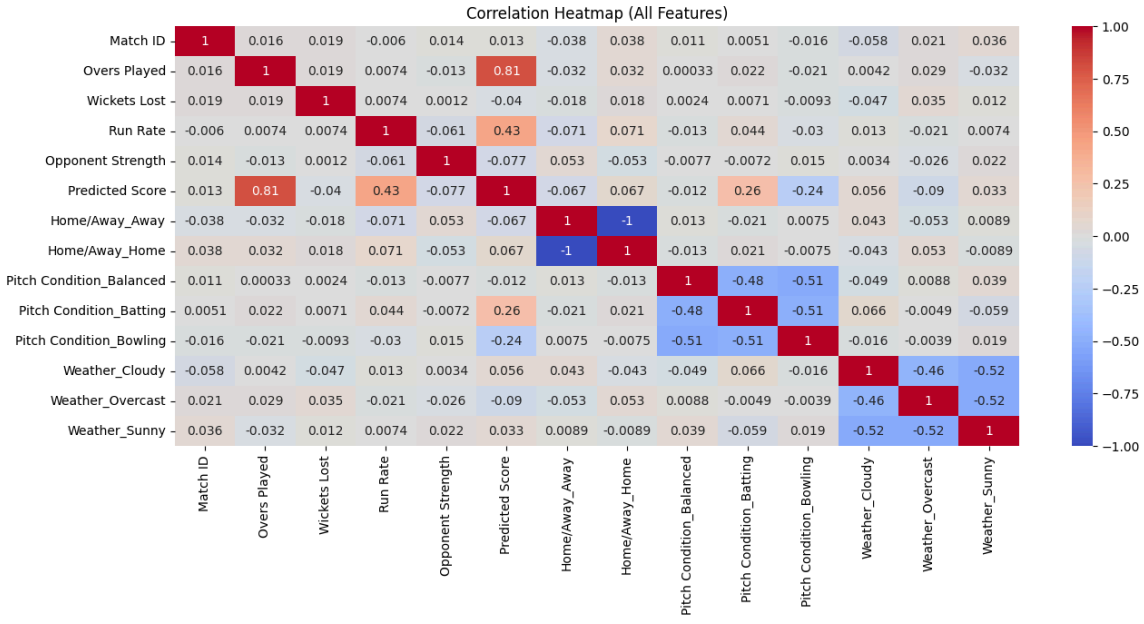
Dataset Description

- There are 9 features in the dataset where “Predicted Score'’ is the target variable.
- It’s a regression problem because the target variable has continuous numerical values.
- The dataset contains approximately 1500 data points.
- There are both categorical and quantitative features in the dataset.
    - ➔ Categorical Features: 'Home/Away', 'Pitch Condition', 'Weather'
    - ➔ Quantitative Features: 'Match ID', 'Overs Played', 'Wickets Lost', 'Run Rate', 'Opponent Strength', 'Predicted Score'
- Yes, we need to encode the categorical features. Because, the machine learning algorithms cannot directly work with string values and require numerical values to work with.

- Correlation Heatmap (Before Encoding)



Correlation Heatmap (After Encoding)



- From the correlation heatmap before encoding (numerical features only), we see that most of the numerical features in the dataset show weak correlations with each other. This indicates that each feature contributes independently to the prediction of the target

variable. However, Overs Played and Run Rate show a relatively stronger positive correlation with the Predicted Score. This suggests that these features have a more direct influence on the final score. For the heatmap after encoding (all features), we see there's little to no change in the numerical features, whereas the categorical feature, 'Home vs Away' has strong negative correlation (-1) with each other. This indicates that match progression related features have a more direct and significant influence on the predicted score compared to other factors such as pitch conditions, weather, or home/away status.

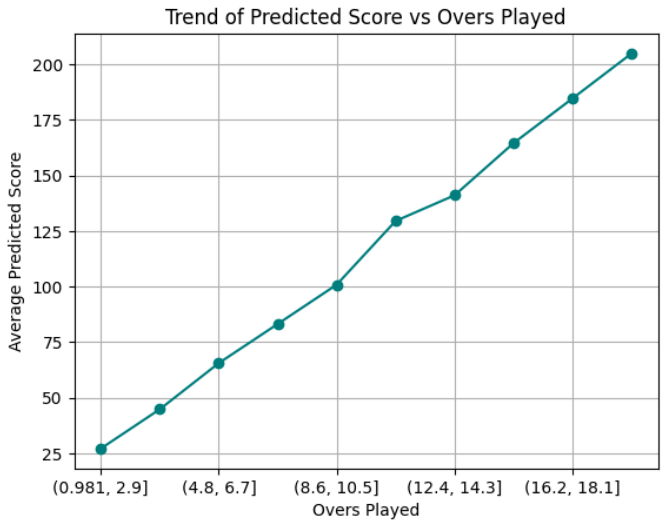
- Since this project deals with a regression problem, the output variable (Predicted Score) is continuous rather than divided into distinct categories. Because of this, a bar chart is not suitable for showing the distribution. Instead, we used a histogram to better represent how the scores are spread across different ranges.

  - From the histogram, it can be seen that most of the scores lie between 20 and 100. Scores above 240 appear much less frequently. This suggests that the data is concentrated around the mid-range values, with fewer observations at the extreme ends. Also the dataset is moderately skewed towards the right side (skewness = 0.877).

- Some important relationships extracted using Exploratory Data Analysis (EDA):

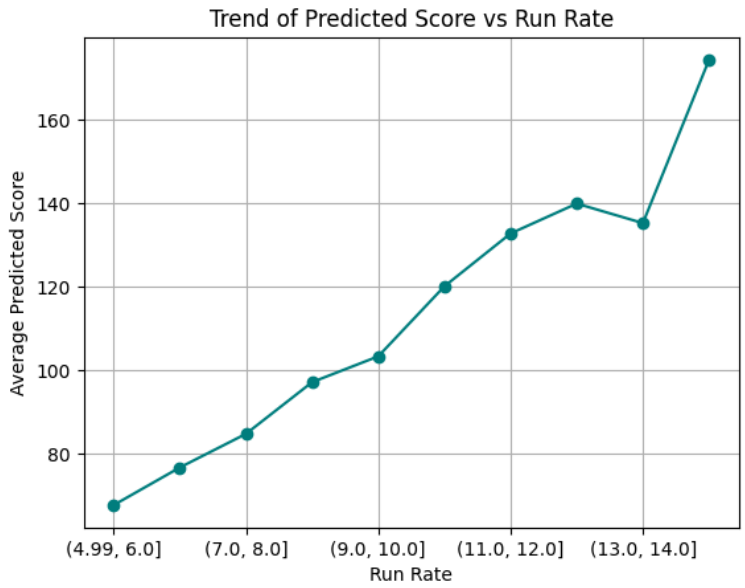
  ➔ <u>Overs Played vs Predicted Score:</u>

  Matches where more overs were played usually had higher predicted scores. This shows a clear positive relationship, since playing more overs gives teams more chances to score runs.



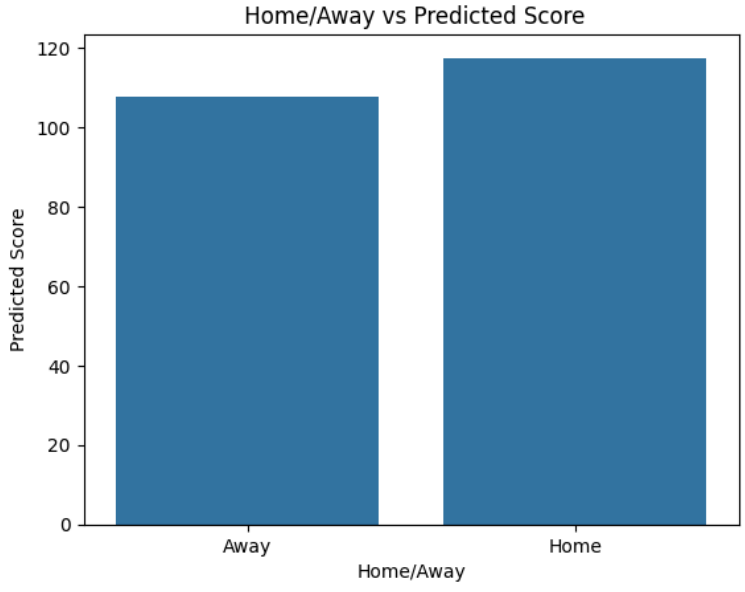
➔ Run Rate vs Predicted Score:

A higher run rate is linked to higher predicted scores. This means that keeping a good scoring pace during the match has a big effect on the final score.



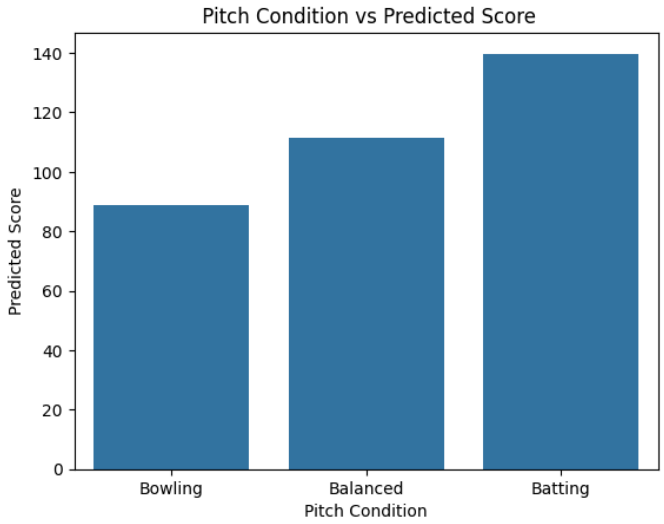
➔ Home/Away vs Predicted Score:

From the bar plots, we can see that match conditions like playing at home or away, as well as pitch and weather conditions, affect the average predicted score. For instance, matches played at home generally had slightly higher average scores than away matches.



➔ Pitch Condition vs Predicted Score:

The bar plot shows that average predicted scores vary depending on the pitch condition. Matches played on batting-friendly pitches tend to have higher predicted scores, while bowling friendly or neutral pitches result in comparatively lower scores. This indicates that pitch condition is an important factor influencing team performance and overall scoring outcomes.



Dataset pre-processing

- Handling Null Values:
  - ➔ Fault: Machine learning models cannot directly process missing values. If null values exist, it is not possible to encode or scale the dataset properly.
  - ➔ Solution: We checked the dataset for missing values and found none. Since the data was already clean, there was no need to remove any rows or columns or apply any imputation. Both numerical and categorical features had no missing values, so no mean or mode replacement was required.

- Encoding:
  - ➔ Fault: Machine learning models can only work with numbers, not strings.

➔ Solution: As the categorical datas are without hierarchy, we used the One Hot method to let the algorithm automatically split the “Home/Away” column into 2 different columns Home/Away_Away and Home/Away_Home, Pitch Condition column into 3 different columns Pitch Condition_Balanced,Pitch Condition_Batting and Pitch Condition_Bowling and Weather column into 3 different columns Weather_Cloudy,Weather_Overcast and Weather_Sunny

- Feature Scaling:
  - ➔ Fault: Machine learning models will generally think features with bigger numbers are more important, which creates bias and inconsistency in result.
  - ➔ Solution: We applied Min-Max Scaler to all numerical features, which scales the data to a specific range (usually 0 to 1). This ensures that all numeric variables contribute equally to the model’s learning process and prevents features with larger values from dominating those with smaller ranges.

Dataset Splitting

We used 80% of the data for training and 20% for testing, and the split was done using a fixed random state (42) to maintain consistency across runs. Since the target variable (Predicted Score) is continuous that’s why stratification was not required.

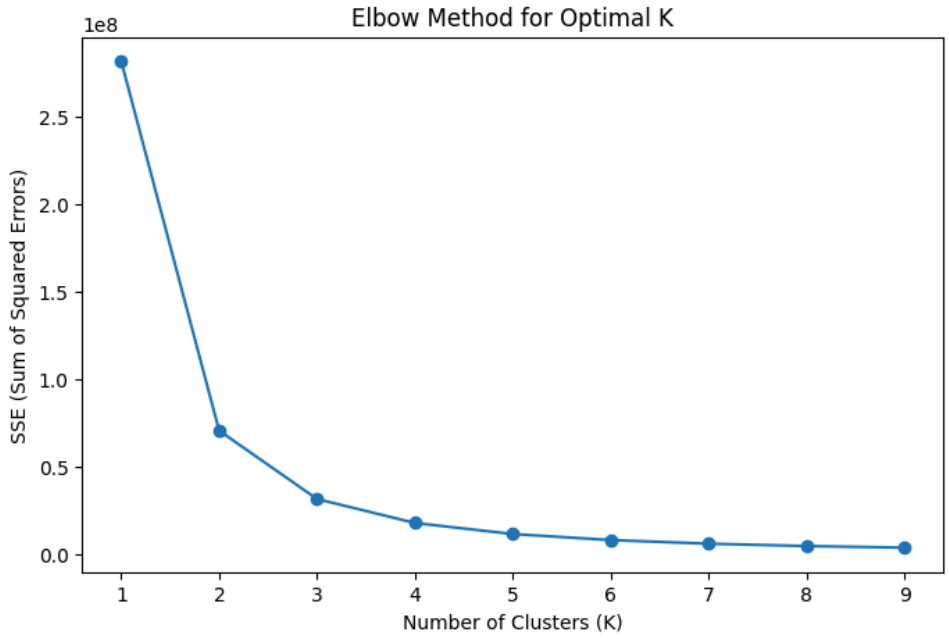
Model Training & Testing

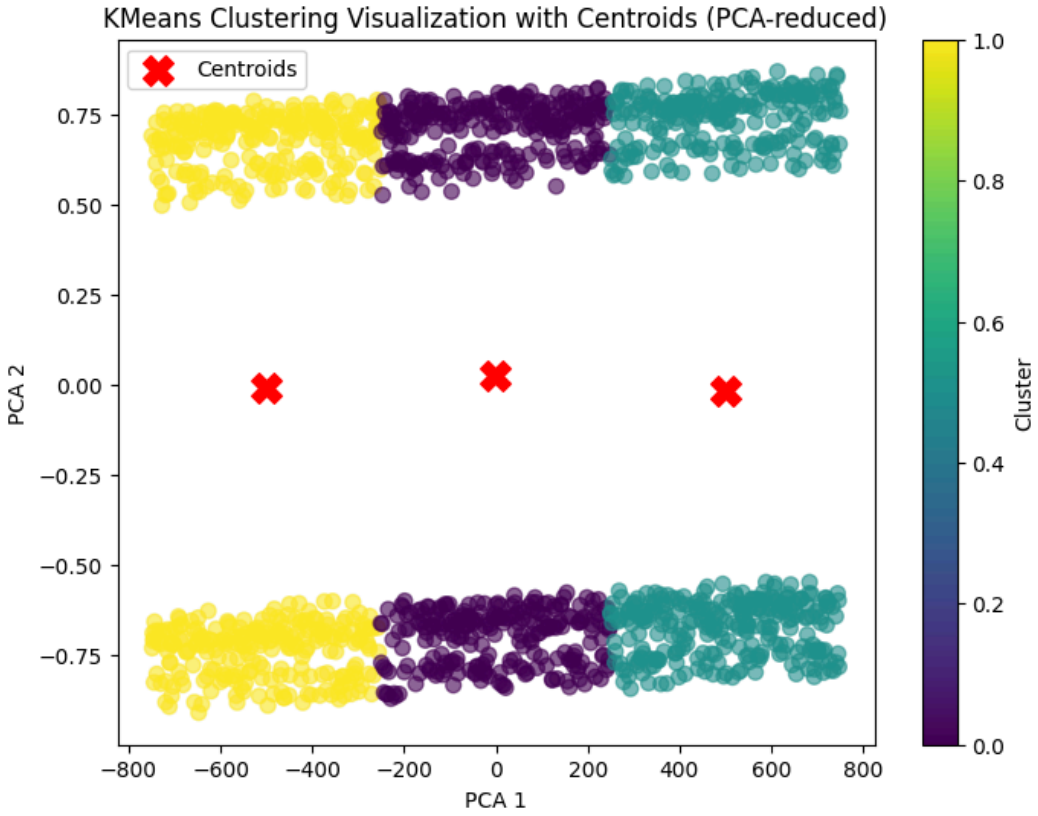
As we have a regression problem, for supervised model training, we used:

1. Linear Regression
2. Decision Tree
3. Neural Network

And, Kmeans Clustering for unsupervised as described.

This is the elbow method:





Comparison Analysis

On the held out test set, the Linear Regression model achieved an R^2 score of 0.913 and an RMSE of 21.010. This shows that while it captured the general scoring trend, it struggled with non-linear relationships and produced larger errors for extreme values.

The Decision Tree Regressor achieved the best overall performance, with an R^2 score of 0.958 and the lowest RMSE of 14.754. This demonstrates that the Decision Tree model effectively captures non-linear patterns in the data while generalizing well to unseen matches.

The Neural Network (MLP Regressor) achieved an R^2 of 0.950 and RMSE of 15.922. Its performance was solid and showed that it could handle some of the non-linear relationships in the data better than simpler models, though it required more data to generalize fully.

Overall, the results indicate that the Decision Tree Regressor is the most suitable model for cricket score prediction in this project, offering the highest accuracy, while the Neural Network serves as a strong alternative and Linear Regression as a reliable baseline.

Conclusion

- The results show that each regression model performed differently. Linear Regression worked as a basic baseline, the Decision Tree performed with the highest accuracy, and the Neural Network showed promise but needed more data to perform well. KMeans clustering grouped matches into general scoring ranges, but it did not clearly separate very high or very low scores.
- Overall, the models were able to capture general patterns, such as the impact of overs played and run rate on the final score. Moreover, the prediction accuracy was fairly high. The Neural Network handled complex relationships slightly better, while the Decision Tree gave inconsistent results on unseen data.
- These results are likely due to noise and variation in the dataset. Some features had a strong effect on the target, while others had very little impact. The score distribution was

also moderately skewed, making extreme values harder to predict. In addition, the dataset size was relatively small, which limited the Neural Network's performance.

- Some of the main challenges were handling categorical features correctly and managing the skewed score distribution. Making sure that preprocessing steps like encoding and scaling were applied properly was also important for getting reliable results.